

Raj Joshi

Software Engineer | Full-Stack (Java, React) | Finance & AI

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PROFILE

Full-stack engineer with 5 years at J.P. Morgan owning Java/Spring Boot and React platforms in asset management, partnering with product owners and operations teams, and mentoring juniors. Daily user of AI-assisted development; built AI agents on AWS Bedrock and Google Vertex AI in hackathons and AWS AI League. Passionate about computer science, finance, and game theory; a curious generalist who learns new domains fast, which matters more in the age of AI.

EXPERIENCE

JPMorgan Chase & Co. | Mumbai

Jul 2021 – Present

Associate, Software Engineer II: Portfolio Insights (Asset & Wealth Management) May 2025 – Present

- Own full-stack features end to end for the **J.P. Morgan Model Portfolios** platform used by **financial advisors**; work with the product owner on discovery, partner with other developers on tight timelines, and review code.
- Extended models to support **alternatives (ALT)** and **separately managed accounts (SMA)**, enabling a business line expected to scale rapidly.
- Integrated Model Portfolios with the **Heatmap** fund-comparison app (incl. capital gains) for composition analysis.
- Built a **white-labeling and co-branding** feature so advisors' firm name and logo appear on every model's trade-commentary and fact-sheet PDF.
- Designed a **fund-suitability** feature (correlation vs. existing portfolio) and single-click portfolio analysis; delivered a **zero-downtime blue/green EKS and EMR migration**.
- Daily **GitHub Copilot** and **Roo Code** user with reusable skills; mentor juniors. Stack: Java, Spring Boot, React, EKS/ECS, Kafka, DynamoDB, Redis.

Analyst & Associate, Software Engineer II: Index Research Jul 2021 – Apr 2025

- **Owned and maintained** a client-reporting application with critical daily-operations modules, including production support, security, and tech-debt remediation; worked with the **operations team** to gather requirements and build solutions for their BAU tasks.
- **Redesigned the database schema** that limited scale, plus caching and backend optimization, so new clients could be onboarded easily and reports delivered reliably to a large client base every month.
- Led the **Java modernization (8 → 11 → 17)** and iBatis → MyBatis migration; removed an outdated Sybase package for faster, more secure releases.
- Delivered end-user feedback full stack (incl. **Lucene** search); built a **Cypress/Cucumber BDD** PoC; contributed to a new **ETF platform** (FastAPI, PostgreSQL, AWS) replacing a legacy system.

PROJECTS

Prayog: Simulated Stock Exchange & Bot Arena • github.com/rj8228/prayog

2026

- Built a single-threaded **matching engine** (Java 21, LMAX Disruptor) with journaling, deterministic replay and snapshots: **~100 ns per order**, **20,000 orders/s** sustained with fsync; 23 architecture decision records.
- Streamed events through **Kafka** into an idempotent post-trade ledger (PostgreSQL, Redis) with P&L and a leaderboard; traced a p99 latency tail to GC pauses with JFR and Prometheus, cutting allocation per order **12–21%**.

Personal Financial Intelligence Agent: Google Cloud Agentic AI Day (Hack2Skill)

Jul 2025

- Shortlisted in the **Top 700 teams of 57,000+** developers. Built an agent with **Gemini, Google ADK, and Vertex AI**, using Fi Money's **MCP** server to deliver personalized financial insights.

AWS AI League: City Lead & SME, J.P. Morgan Mumbai Tech Center

2025 – 2026

- Ran the site program for AWS's gamified agentic-AI competition: participants built **multi-agent systems on Amazon Bedrock (Agents, Lambda)** to learn prompt engineering and agent design.

SKILLS

Backend: Java, Spring Boot, MyBatis, Lucene, Python, FastAPI, Django, Kafka, REST

Frontend: React, Redux, Vite, JavaScript

Data & Cloud: AWS (EKS, ECS, EMR, Lambda, Bedrock), Docker, DynamoDB, Redis, PostgreSQL, Terraform

Engineering: Jenkins, Git, Sonar, Cypress/Cucumber BDD, ServiceNow, CI/CD

AI: LLM agents (Bedrock, Vertex AI, ADK), MCP, context and prompt engineering, agent skills, GitHub Copilot, Roo Code

LEADERSHIP & RECOGNITION

- **Co-led** the DeepRacer competition for **700+** participants (2024); **led a team of 6 to 2nd place** (2023)
- **Winner**, SEPathon hackathon (2022) and Induction Program (2021), built an algorithmic-trading training app